

Luis San Martin

Computer Vision Engineer · Software Developer

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PROFILE

Over three years building software that stays running in production. I train and deploy **computer vision models** — object detection, classification and tracking — on cameras in real operation, and build the data systems that integrate and report that information unattended. Training runs in the cloud; inference stays on the client's own hardware. I also leave **agents running on their own** — reports, alerts and monthly indicators over messaging — bringing in language models where they help and leaving them out where they are not needed. I work directly with the owner of each operation, from the requirement to the running system. Automation Engineering, Universidad del Bio-Bio.

TECHNICAL SKILLS

Computer vision	PyTorch · YOLOv8 · OpenCV · object detection and classification · multi-object tracking (SORT) · <i>fine-tuning</i> on custom datasets · in-field image annotation · real-time inference
Training & deployment	Cloud training (Google Colab) · local <i>on-premise</i> inference on the client's GPU · edge deployment (ESP32-S3) · IP cameras over RTSP · evaluation by mAP, precision and recall
Agents & generative AI	Conversational agents on WhatsApp and Telegram in operation · LLM API integration (Groq, Anthropic) · intent classification · custom n-gram classifier with no API dependency · LLM observability with Langfuse and OpenTelemetry
Languages	Python · SQL · TypeScript / JavaScript
Data & reporting	PostgreSQL (modelling, versioned migrations, row-level security) · Firebird · SQLite · ETL and integration of systems with no API · web scraping · real-time dashboards · automated reports and threshold alerts
Infrastructure	Linux · Docker · Git · GitHub Actions · cloud servers · scheduled jobs and monitoring

EXPERIENCE

Software Developer — *Distribuidora El Maravilloso*

Nov 2025 – Present · Concepcion, Chile

- Integrated the point-of-sale system — a Firebird database with **no API** — into a cloud PostgreSQL database: mapped the source schema and built the synchronisation, today holding **11,900 sales and 40,000 product lines** (~78 sales per day).
- **Unattended ETL** that pulls tax documents from the national tax authority portal and computes the monthly tax indicators: **305 purchase invoices** imported and classified across 87 suppliers.
- **Data quality control:** cross-checking three sources — tax documents, electronic payments and point-of-sale records — to detect duplicated income and till mismatches: **283 alerts** raised without altering the client's figures.
- **Agents in operation** over WhatsApp and Telegram: they deliver the sales close and the month's tax indicators, and raise threshold alerts when something does not reconcile. Built a **custom n-gram intent classifier** (82% accuracy, typo-tolerant) that resolves classification **without calling any API**, and instrumented the AI routes with Langfuse.
- Real-time reporting, monthly executive report, and diagnosis of **data integrity incidents in production** (record loss from an over-broad cleanup rule, clock drift between source and cloud), recovering the affected data.

Stack: Python · SQL · PostgreSQL · Firebird · JavaScript · Linux · scheduled jobs

- Migrated the business off the commercial platform they were using — **~170 active members** —: extracted their full database by **web scraping**, normalised it and moved clients and history across with no loss.
 - Designed the **PostgreSQL** schema: multi-site with row-level security and versioned migrations, for **2 locations**.
 - Built the admin panel and its reporting: attendance, occupancy by site and time slot, revenue, and commission settlement for **14 professionals** across a 768-session schedule.
- Stack: PostgreSQL · SQL · Python · TypeScript

Computer Vision Engineer — St. Andrews · Mussels Chiloe

Jun 2024 — Jul 2025 · Puerto Montt, Chile

- Trained and deployed a multi-camera **detection, classification and counting** system across **3 packing stations** on a line handling **13,000 kg per day**, at the world's largest mussel producer.
 - **Multi-object tracking (SORT)** to hold each package's identifier across frames and prevent double counting when the belt stops or an operator reaches in.
 - **Larvae counting model** that cut manual analysis from **45 minutes to seconds**: 696 individuals in 2.93 s at 85% accuracy, trained on a **custom dataset annotated inside the plant**.
 - Robust to changing lighting, moving cameras and wet surfaces. **Ran a full season with no manual intervention**, with inference running on the plant's own machines.
- Stack: Python · PyTorch · YOLOv8 · OpenCV · SORT · RTSP over IP cameras

COMPUTER VISION & AI PROJECTS

Helmet detection on an embedded camera — Ewood · electric mobility

2026

- **Fine-tuned** YOLOv8 to 3 classes (scooter, helmet, no-helmet) over **12,321 images** and 80 epochs: **mAP@50 of 85.5%**, precision 85.0% and recall 80.9%.
 - Live inference at **9.6 frames per second** on an **embedded XIAO ESP32-S3 Sense camera**, not a CCTV camera: the constraint was that the hardware had to be mountable on the scooter itself.
- Stack: PyTorch · YOLOv8 · OpenCV · Python · ESP32-S3

Retail footfall analytics — with client, in production

Feb 2026 — Present

- Entry and exit counting on the CCTV cameras the store already had, cumulative traffic heatmaps and configurable line-crossing alerts, with no additional hardware.
- Stack: Python · YOLOv8 · OpenCV

Personal protective equipment (PPE) detection — final-year project

2026 — ongoing

- Detection of helmets, vests and other PPE on industrial sites. The core contribution is the **custom dataset annotated on a real work site**, rather than public banks that do not represent actual working conditions.
- Stack: PyTorch · YOLOv8

EDUCATION

Automation Engineering — Universidad del Bio-Bio

Graduated 2024 · Degree completion 2026

Languages	Spanish (native) · English (technical reading)
Availability	Immediate · Concepcion, Chile — remote or hybrid

Case studies with images and metrics at [sanmaglass.github.io](#). Repositories are private as they hold client production code; available for review under agreement.